Wireless Security

1. Introduction

Wireless Security refers to the protection of wireless networks (Wi-Fi, Bluetooth, etc.) from unauthorized access, attacks, and data theft.

Unlike wired networks, where physical access is required, **wireless networks transmit data over radio waves**, making them more **vulnerable to interception and attacks**.

Wireless security ensures that **only authorized users** can connect, and that **data transmitted** over wireless links remains **confidential and intact**.

2. Objectives of Wireless Security

1. **Confidentiality** – Protect data from being intercepted or read by unauthorized users.
2. **Authentication** – Verify the identity of devices before allowing access.
3. **Integrity** – Ensure data isn't modified during transmission.
4. **Availability** – Keep the wireless network accessible and operational.
5. **Access Control** – Restrict unauthorized devices from joining the network.

3. Common Threats in Wireless Networks

Threat	Description
Eavesdropping	Attackers capture wireless signals to read confidential data.
Rogue Access Points	Unauthorized Wi-Fi hotspots used to trick users into connecting.
Man-in-the-Middle Attack (MITM)	Attacker intercepts communication between two parties.
Denial of Service (DoS)	Network is flooded with traffic to make it unavailable.
MAC Address Spoofing	Attacker changes their device's MAC address to bypass filters.
Replay Attacks	Captured data packets are resent to gain unauthorized access.

4. Wireless Security Protocols (Wi-Fi Security Standards)

Over time, multiple security standards have been developed for Wi-Fi encryption.

Protocol	Full Form	Encryption Used	Security Level	Remarks
WEP	Wired Equivalent Privacy	RC4	Weak	Easily hacked; outdated
WPA	Wi-Fi Protected Access	TKIP	Moderate	Temporary improvement over WEP
WPA2	Wi-Fi Protected Access 2	AES	Strong	Widely used and secure
WPA3	Wi-Fi Protected Access 3	SAE (Simultaneous Authentication of Equals)	Very Strong	Latest standard; resists password guessing

5. Key Security Mechanisms

1. **Encryption:**

Converts readable data into ciphertext to prevent eavesdropping.

   a. Protocols: AES (used in WPA2/WPA3)

2. **Authentication:**

Ensures only legitimate users connect to the network.

   a. Example: WPA2-Enterprise uses **RADIUS server** for user verification.

3. **Access Control:**

Uses **MAC filtering**, **firewalls**, and **VPNs** to limit access.

4. **Firewalls and Intrusion Detection Systems (IDS):**

Monitors and filters malicious wireless traffic.

5. **Network Segmentation:**

Keeps guest Wi-Fi and internal networks separate for safety.

6. Best Practices for Wireless Security

- Use **WPA3** or at least **WPA2** encryption.
- Change default router passwords.
- Use **VPNs** on public Wi-Fi.
- Regularly update router firmware.
- Disable **WPS (Wi-Fi Protected Setup)** feature (it can be exploited).

7. Wireless Security in Different Technologies

Technology	Security Features
Wi-Fi (802.11)	WPA2/WPA3 encryption, authentication mechanisms
Bluetooth	Pairing process, PIN verification, short-range communication
Mobile Networks (4G/5G)	SIM-based authentication, encryption at radio layer
IoT Wireless Devices	Lightweight encryption (AES-128), secure boot, and access control

8. Tools for Wireless Security

Tool	Purpose
Aircrack-ng	Wi-Fi password cracking and testing encryption strength
Wireshark	Network packet analysis
Kismet	Wireless network detection and intrusion detection
NetStumbler	Detects wireless networks and access points
Nmap	Scans and audits network security

9. Advantages of Wireless Security

- Protects data from eavesdropping.
- Prevents unauthorized access.
- Increases user trust and privacy.
- Reduces risk of data breaches.
- Ensures safe communication in open/public Wi-Fi areas.

10. Limitations of Wireless Security

- Encryption can slightly reduce network speed.
- Outdated routers may not support modern protocols like WPA3.
- Misconfiguration (weak passwords, open networks) still makes systems vulnerable.
- Rogue access points can still deceive users in public areas.